

customers about tick sizes.¹³⁹³ For these reasons, the half-penny tick is not expected to disadvantage retail traders.

One commenter requested that the Commission consider how a smaller tick size could affect stock splits.¹³⁹⁴ The commenter cited academic research suggesting that stock splits can be used to affect the bid-ask spread.¹³⁹⁵ To the extent that issuers engage in stock splits to manage their quoted spread, this behavior is likely to continue under the amended rules when issuers believe that a stock split could improve their liquidity. However, the amendments are expected to improve liquidity on average for stocks subject to the smaller tick size, so it may be less likely that an issuer may feel the need to manage liquidity via stock splits going forward and so there could be fewer associated stock splits for this reason. Furthermore, although stock splits and reverse splits can mechanically affect the bid-ask quoted spread and change the optimal tick, the next evaluation period would rectify any misalignment in the stock's tick size assignment.¹³⁹⁶

Some commenters requested that the Commission provide an analysis of the effect of the amendments on the use of ISOs (intermarket sweep orders).¹³⁹⁷ Some stated that spreading liquidity over more price levels would increase the risk of information leakage with regards to a large order being split over many smaller orders which would lead to an increased complexity of implementing large orders and would thus lead to an increased use of ISO orders.¹³⁹⁸ Barardehi

¹³⁹³ See supra note 307 and surrounding text.

¹³⁹⁴ See CCMR Letter at 28-29.

¹³⁹⁵ See James J. Angel, Tick Size, Share Prices and Stock Splits, 52 J. FIN. 655 (1997), and Sida Li & Mao Ye, supra note 1286.

¹³⁹⁶ See supra section VII.C.1.c for additional discussion of tick sizes, quoted spreads, and stock splits. See supra note 1367 and supra section III.C.8 for discussions of how tick sizes are assigned following a stock split or reverse split.

¹³⁹⁷ See SIFMA Letter II at 34. See also 17 CFR 242.600(b)(47) (defining an intermarket sweep order).

¹³⁹⁸ See, e.g., Vanguard Letter at 5, Citigroup Letter at 4, and Virtu Letter II at 17.

et al. (2022)¹³⁹⁹ specifically examine the effect of tick sizes on ISO activity. Their study finds that a narrower tick is associated with more ISO activity which the authors attribute to the increased market complexity associated with implementing trades in a narrower tick environment. This is consistent with the effect considered by the commenters. Consequently, ISO usage is likely to increase for stocks that receive the narrower tick size. One commenter stated that increased use of ISO orders may result in more locked and crossed markets.¹⁴⁰⁰ Research on the link between ISO usage and locked markets is scant. Increased ISO usage could eventually lead to an increase in locked and crossed markets, which could make transacting on exchanges more complicated, however the magnitude of any effect is uncertain.

Another commenter specifically asked the Commission to consider the effect of a lower tick size on ETFs as opposed to stocks.¹⁴⁰¹ The commenter stated that ETFs tend to have larger trade sizes and also that the creation and redemption process for ETFs is unique. However, the commenter does not provide any analysis regarding how these differences would lead an ETF to react differently than a common stock to a reduction in the tick size.¹⁴⁰² The fundamental economics regarding the tick size tradeoff discussed at the beginning of this section applies to ETFs because the economics discussed in this section rely on the mechanics of quoting and trading, not on the assets underlying the stock or ETF. It is also unclear how the creation redemption process would differ from the analysis provided above. Additionally, the creation/redemption process also does not produce unique economics in the context of these

¹³⁹⁹ See Barardehi et al., supra note 231.

¹⁴⁰⁰ See RBC Letter at 3.

¹⁴⁰¹ See Tradeweb Letter at 3.

¹⁴⁰² Id.

amendments. This is because authorized participants purchasing the underlying shares to deliver in exchange for shares of the ETF or delivering shares of the ETF in exchange for the underlying assets would still need to purchase and sell the underlying shares in the stock market, subjecting them to the economics of supply and demand for liquidity provision for the stocks in question. Consequently, ETFs with narrower quoted spreads likely will experience an improvement in market quality with a \$0.005 tick size. To the extent that ETFs have larger average trade sizes the benefits of the Rule may be smaller, consistent with the analysis presented in Barardehi et al. (2022). But as the adopted amendment is more conservative than the proposed rule in that it applies a tick size of \$0.005 to stocks with quoted spreads equal to or less than \$0.015, the Commission does not believe, based on the analysis above and commenters evidence presented above, that the amendments are likely to harm ETFs that receive a tick size reduction.

One commenter stated that updated Rule 605 data would “question the validity of assumptions” made in the tick size proposal.¹⁴⁰³ The commenter stated that Rule 605 execution quality data for stocks that have quoted spreads wider than the tick would demonstrate that the minimum quoting increment is not the driver of off-exchange retail trading.¹⁴⁰⁴ The Commission agrees that there are other factors driving off-exchange retail trading, and adopts changes to Rule 612 for reasons other than a significant change in retail order flow. For this reason, additional information regarding retail execution quality that will arise from amended Rule 605 is not needed prior to adopting amendments to Rule 612.

One commenter stated that variable tick sizes “raise[] concerns about the ability to compare the execution quality for the stock across multiple months,” resulting in “a significant

¹⁴⁰³ See Citadel Letter III at 1-2.

¹⁴⁰⁴ Id.

possibility of investor confusion when comparing Rule 605 reports across several months.”¹⁴⁰⁵

The Commission acknowledges that changes in the tick size may result in changes to the levels of some measures of execution quality that are sensitive to the tick size, such as price improvement, over time. To the extent that this reduces the interpretability of Rule 605 reports, particularly for stocks that experience frequent changes in the tick size, this could represent a cost of the amendments. However, there are several factors that will mitigate this potential cost. First, the adopted amendments include an operative period that limits the frequency at which a tick assignment is updated to a minimum of six months. The fact that a stock’s tick assignment cannot vary more frequently than every six months greatly reduces the number of potential changes in execution quality levels in monthly Rule 605 reports that result from changes to a stock’s tick size. Second, to the extent that market participants will be able to combine Rule 605 information with information about a stock’s historical tick size,¹⁴⁰⁶ this will allow them to control for this characteristic when assessing a stock’s execution quality data over time. In addition, as acknowledged by the commenter, a change in the tick size “may impact market centers and broker-dealers reporting under 605 in the same manner,”¹⁴⁰⁷ such that variations in the tick size (and the resulting mechanical effects on execution quality levels) will not impact the use of Rule 605 to compare execution quality across reporting entities within a given month.

If FINRA chooses not to update Rule 5320 (the ‘Manning Rule’), the lower tick size could make claiming the price improvement exception to FINRA Rule 5320 harder for market

¹⁴⁰⁵ See SIFMA Letter II at 20-21.

¹⁴⁰⁶ While there is no requirement for the listing exchange to disseminate historical information about the tick size, it is likely that this information will be collected and disseminated by other market participants, such as firms providing services related to financial data and analysis because there would likely be demand for such data because, for example, it would be needed to perform after the fact transaction cost analysis.

¹⁴⁰⁷ See SIFMA Letter II.

participants since it would require a two-tick price improvement for some stocks instead of a one tick price improvement. One commenter stated that because the price improvement exception in the Manning Rule is currently tied to \$0.01 price improvement, a tick size smaller than \$0.01 would “greatly increase the cost and complexity of compliance and would likely disincentivize (or eliminate) the handling of customer limit orders by wholesale broker dealers.”¹⁴⁰⁸ The commenter made this statement specifically referencing the proposed \$0.001 tick increment. The adopted amendments do not include the \$0.001 tick size and so these concerns are significantly mitigated. Nonetheless, requiring two tick price improvement for some orders could increase the complexity associated with complying with the Manning Rule, particularly in situations where the quoted spread is only one tick wide because it would require more than crossing the quoted spread in order to claim the exception. For broker-dealers, such as wholesalers, whose business models center on internalizing customer orders within the NBBO, the requirement to, in some instances, more than cross the quoted spread in order to execute a customer order could be a disincentive to handling some orders as it could render such trades unprofitable.¹⁴⁰⁹

One commenter stated that a lower tick size could lead to oscillation in some stocks between tick sizes.¹⁴¹⁰ The commenter stated that a stock that falls just under the threshold and thus receives a smaller tick may be subject to more undercutting with the smaller tick size, which could cause quoted spreads to widen. Wider spreads would make the stock revert to the wider tick size, which would reduce undercutting so that quoted spreads would decline. A narrower

¹⁴⁰⁸ See Citadel Letter I at 8.

¹⁴⁰⁹ See supra note 193 and surrounding text for further discussion.

¹⁴¹⁰ See Mitre Corp. Letter at 5.

spread could lead to a smaller tick in the next round and so on.¹⁴¹¹ The Commission believes that this outcome is unlikely given the analysis provided in this section. Stocks receiving the smaller tick size are expected to experience smaller quoted spreads due to the smaller tick size allowing pricing that better reflects supply and demand. This effect would reduce oscillation. Stocks receiving the smaller tick size would likely experience tighter quoted spreads making it less likely that they would revert to the \$0.01 tick in the next evaluation period.

2. Lower Access Fee Cap

The amendments will lower the access fee cap from \$0.003 per share (30 mils) to \$0.001 per share (10 mils) for NMS stocks priced at \$1.00 or more, and from 0.3% to 0.1% of the share price for stocks with prices less than \$1.00. Lowering the access fee cap preserves price coherence,¹⁴¹² given changes to the tick size. Moreover, the Commission expects that lowering the access fee cap will result in lower transaction costs for investors. The Commission also expects that a lower access fee cap will result in wider quoted spreads; however, market quality will nonetheless improve. Lowering the access fee cap will reduce exchange transaction revenue due to lower capture on sub-\$1.00 stocks. We describe these effects in more detail below.

Commenters, with few exceptions,¹⁴¹³ agreed on the need for Commission action on access fees given the change in the tick size.¹⁴¹⁴ One commenter stated that in light of the

¹⁴¹¹ Id.

¹⁴¹² See infra section VII.D.2.a

¹⁴¹³ Exceptions include e.g. Citigroup Letter at 6, and World Federation of Exchanges Letter at 4, Pragma Letter at 7, Hudson River Letter at 4, and Budish Letter at 6-7. However, these commenters did not present arguments suggesting that an access fee greater than 50% of the tick size would not cause price coherence problems. The Commission believes that retaining a 30 mil access fee for stocks trading with a \$0.005 tick would further separate the price from the economics of the trade and disrupt the coherence between nominal and net pricing as the access fee cap would be greater than 50% of the tick size.

¹⁴¹⁴ See, e.g., Nasdaq Letter I at 19 (stating “Nasdaq recognizes that if Commission action successfully updates tick sizes and narrows spreads for certain stocks, then existing exchange access fees and rebates may no

reduction in ticks for some stocks to \$0.005, leaving the access fee cap at 30 mils would “distort trading economics in a manner that undermines the Commission’s goals for competition and Best Execution.”¹⁴¹⁵ Many commenters supported a 10 mils access fee.¹⁴¹⁶ Some commenters went further, stating that the Commission should explore “comprehensive access fee reform” or ban rebates entirely.¹⁴¹⁷ Some commenters suggested what they viewed as a less extensive change, namely an alternative in which some stocks had a fee of 15 mils whereas others had a fee of 30 mils.¹⁴¹⁸ In short, while commenters agreed on the need for Commission action to lower the access fee cap, they disagreed regarding the specifics.

a. Coherence Between Net and Quoted Prices

In the Proposing Release, the Commission discussed the need to maintain an access fee cap that is less than half of the tick size due to the need to maintain coherence between net and

longer be appropriate.”), See Cboe, State Street, et al. Letter at 3 (stating “We acknowledge that a reduction in quoting increments for tick constrained symbols could make it advisable for market centers to reduce access fees for the affected symbols to ensure a consistent equity market structure framework.”), see Better Markets Letter II at 3-4 (stating “A reduction in the minimum tick size without reducing access fees could permit fees to become a higher percentage of the minimum pricing increment, which would almost certainly undermine price transparency.”), and see RBC Letter at 4 (stating “If the MPIs are meaningfully reduced as noted in the Proposal, then access fees would need to be lower as well.”).

¹⁴¹⁵ Nasdaq Letter I at 19.

¹⁴¹⁶ See, e.g., BlackRock Letter at 10-11, BMO Letter at 3-4, Budish Letter, IEX Letters I-V, JPMorgan Letter at 6, NASAA Letter at 9, Vanguard Letter at 2 and 6, XTX Letter at 5.

¹⁴¹⁷ See We The Investors Letter I at 3-4 (recommending banning rebates); Harris Letter at 4 (recommending reverting to traditional fees, thereby effectively eliminating rebates). BlackRock Letter at 11 states “Although we believe that the current proposal may miss an opportunity to enact more holistic and lasting access fee reform, we concede that, for highly liquid securities, a 10 mil access fee cap reasonably threads the needle between countervailing adverse consequences. Accordingly, under a uniform fee model, we would be supportive of setting the access fee cap at 10 mils.”

¹⁴¹⁸ See infra note 1805 for a list of commenters suggesting this alternative, and a discussion of the costs and benefit of this alternative.

quoted prices.¹⁴¹⁹ Commenters, with few exceptions,¹⁴²⁰ agreed.¹⁴²¹ Reducing the access fee cap to 10 mils will satisfy this condition of coherence. As explained in the Proposing Release, only the best posted price is protected. Under the current regulatory framework, leaving the access fee cap at 30 mils could preclude market participants from trading on exchanges that have the best displayed price when fees and rebates are included. Suppose a traditional exchange has a displayed protected bid at \$10.010, whereas an inverted exchange has a displayed protected bid at \$10.005. Order protection would require the exchange to have policies and procedures in place reasonably designed to prevent trades from occurring at a price worse than the protected quote,¹⁴²² effectively requiring the investor to go to the traditional exchange if the investor wished to trade against a displayed, on exchange bid. However, on the traditional exchange, the investor demanding liquidity would take home \$10.007, whereas on the inverted exchange, the investor would take home \$10.008.¹⁴²³

Closely related to the lack of price coherence on exchanges is the effect on price transparency. To illustrate, consider a situation with an access fee and rebate of \$0.003 and a tick

¹⁴¹⁹ See Proposing Release, *supra* note 11, at 80348 n.712. Net and quoted price rankings are coherent if sorting trading venues on the competitiveness of their quoted prices yields the same ordering as sorting on prices net of fees and rebates.

¹⁴²⁰ Exceptions are Citigroup Letter at 6, and World Federation of Exchanges Letter at 4. These commenters do not present arguments that counter those others in the comment file. The Commission believes that retaining a 30 mil access fee for stocks trading with a \$0.005 tick would further separate the price from the economics of the trade and disrupt the coherence between nominal and net pricing as the access fee cap would be greater than 50% of the tick size.

¹⁴²¹ See, e.g., MEMX Letter at 22-24 and Pragma Letter at 7.

¹⁴²² See 17 CFR 242.611(a)(1); see also 17 CFR 242.611(b) (exceptions).

¹⁴²³ For the liquidity demander in this example, the net proceeds of selling to the liquidity provider on the traditional exchange would be the quoted price of \$10.01 less the \$0.003 access fee, or \$10.007. On the inverted venue the liquidity demander would receive the quoted price of \$10.005 plus a taker rebate of \$0.003 from selling, or \$10.008. Although the liquidity demander would receive a better net price by selling at the inverted venue, because the traditional exchange has the better quoted price the order protection rule will prevent the trader from accessing the liquidity on the inverted venue before first accessing the liquidity on the traditional exchange.

size of \$0.005. Consider the effect on prices of a stock. Suppose one sees a trade executed at a price of \$10.005 followed by another executed at a price of \$10.010. Many investors would interpret this as a sign that a stock was increasing in value. However, with an access fee of \$0.003, the net price of the first order if it represents a market order to buy is \$10.008 (the buyer pays \$10.005 + \$0.003), whereas the net price of the second order if it is a market order to sell, is \$10.007 (the seller receives \$10.010 and pays \$0.003 in fees). The price has fallen, not risen, an effect that only the most sophisticated market participants would be able to discern.¹⁴²⁴

Lowering the access fee cap to 10 mils would solve both of these problems.

b. Quantitative Net Capture Analysis

While the amendments do not directly dictate what rebates trading venues can offer, trading venues generally finance rebates through access fees, so in practice reducing the access fee cap will lower the rebates offered.¹⁴²⁵ If trading venues were to subsidize rebates by taking a net loss per share transacted, they would be vulnerable to experiencing extreme and unpredictable losses if volumes spike. Such a trading venue could experience such losses if its non-transaction fee sources of revenue do not increase enough with a spike in trading volume to offset their negative net capture. Trading volumes can vary significantly through time, with little ability for a trading venue to predict the timing and magnitude of changes in trading volume. For example, in January 2021 volume spiked dramatically for certain stocks relative to pre-January 2021 levels.¹⁴²⁶ Exchanges could face financial hardship should rebates deviate substantially

¹⁴²⁴ See Budish Letter at 6.

¹⁴²⁵ See supra note 1077 and surrounding text discussing that access fees fund transaction rebates and while trading centers could subsidize rebates with non-fee revenues they do not do so in practice.

¹⁴²⁶ See Staff Report on Equity and Options Market Structure Conditions in Early 2021 (Oct. 14, 2021) available at <https://www.sec.gov/files/staff-report-equity-options-market-struction-conditions-early-2021.pdf>.

from fees; so it is unlikely that exchanges would take this risk. For this reason, rebates and the cap on access fees are tied together.

As explained in section VII.C.2.b, the Commission understands that the net capture for non-auction trading in stocks that have a price equal to or greater than \$1.00 is likely close to 2 mils for most exchanges. An exchange net capture rate of approximately 2 mils is in line with current pricing practices at most exchanges; it is reasonable to estimate that exchanges would realize a similar net capture rate because the current net capture rate will remain possible under the adopted amendments. The Commission acknowledges uncertainty over whether this 2 mils capture rate will persist or be different should trading venues choose to alter their business model in response to the change in access fees. The analysis that follows assumes that exchanges will maintain the practice of financing rebates through access fees, and thus for transactions in stocks priced \$1.00 or more the Commission expects the average access fee to be near the 10 mil access fee cap and the average rebate to be approximately 2 mils lower.¹⁴²⁷ The analysis also assumes that the behavior of inverted exchanges and off-exchange venues changes proportionally. Although the amendments would not require proportional change on the part of inverted venues, there is currently no restriction on the level of rebates for taking liquidity or fees for posting; yet, as shown in Table 4 in section VII.C.2, inverted venues generally have fee and rebate levels similar to maker-taker venues and approximately a 2 mils capture rate.

Table 13 uses volume estimates from Table 5 to provide estimates of the fees and rebates that would have been collected and disbursed in 2023 if the amended access fee cap was

¹⁴²⁷ See section VII.C.2 for additional discussion about the roughly 2 mil estimated net capture rate for exchanges. At certain pricing tiers rebates may exceed the access fee cap. However, because total overall fees exceed the total rebates paid out, the average rebate would remain lower than the average access fee.